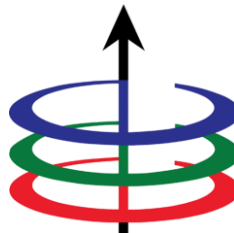


University of the Philippines Diliman
College of Engineering
Electrical and Electronics Engineering Institute

First-Order Circuits Part 2

1st Semester, A.Y. 2026–2027



Student Outcomes

- Recall and apply the DC steady-state characteristics of L and C
- Apply the general step-by-step procedure to analyze first-order circuits and formulate the complete response.
- Analyze circuit behavior under switched and pulsed excitations using the unit step function $u(t)$ and time-shifted pulses.

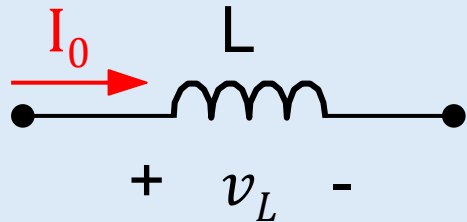
Lecture Outline

- Review of the DC Steady-State Characteristics of L and C
- General Procedure in Solving First-Order Circuits
- The Unit Step Function

Recall

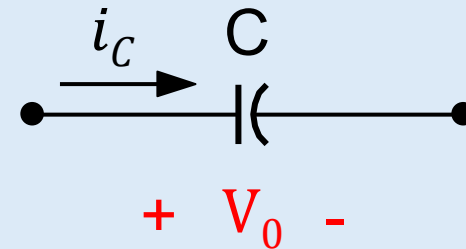
L and C at Steady State

With all **sources constant**, then at steady-state, all currents and voltages are constant.



If the current is constant, then:

$$v_L = L \frac{dI_0}{dt} = 0$$



If the voltage is constant, then:

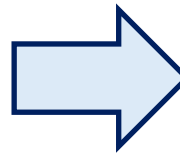
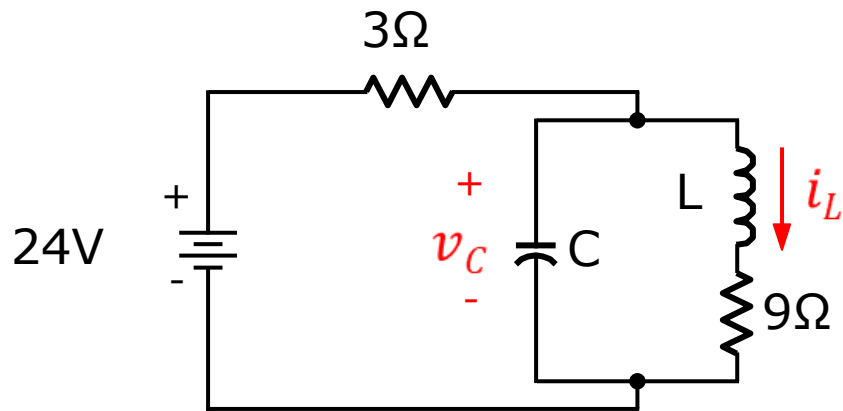
$$i_c = C \frac{dV_0}{dt} = 0$$

Note: With constant sources, L is short-circuited and C is open-circuited at steady state condition.

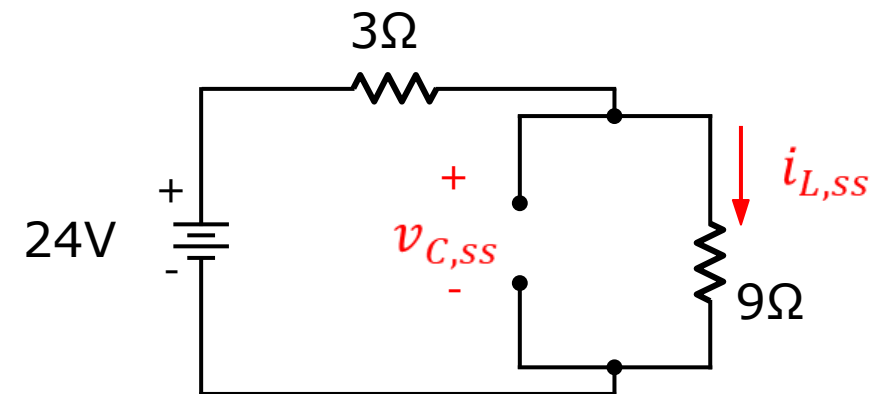
Recall

Example 1:

Find the inductor current and capacitor voltage at steady state.



At steady state, short the inductor and open the capacitor.



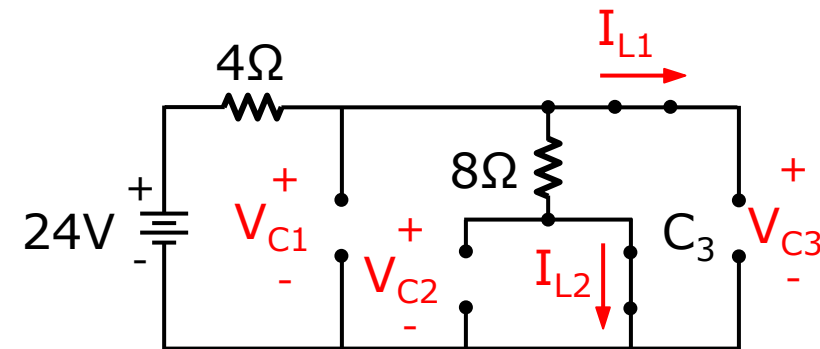
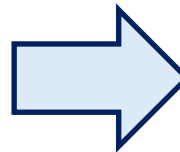
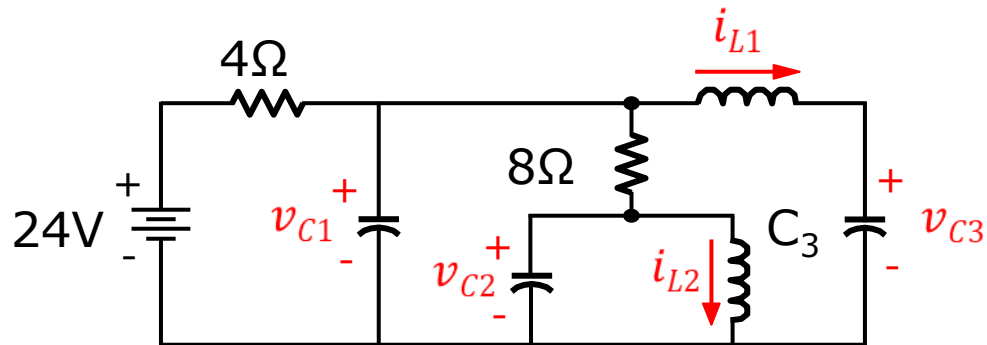
$$i_{L,ss} = \frac{24}{12} = 2 \text{ A}$$
$$v_{C,ss} = 9i_{L,ss} = 18 \text{ V}$$

Recall

Example 2:

Find the inductor currents and capacitor voltages at steady state.

At steady state, short the inductor and open the capacitor.



$I_{L1} = 0$	$V_{C1} = 16 \text{ V}$
$I_{L2} = 2 \text{ A}$	$V_{C2} = 0$
	$V_{C3} = 16 \text{ V}$

General Procedure for First-Order Circuits

- 1. Initial Value:** Determine $f(0^+)$ using continuity conditions:
 - $i_L(0^+) = i_L(0^-)$
 - $v_C(0^+) = v_C(0^-)$
- 2. Final / Steady-State Value:** Determine $f(\infty)$ under DC steady-state conditions:
 - capacitors $\rightarrow$ open
 - inductors $\rightarrow$ short
- 3. Time Constant:** Deactivate independent sources. Simplify the network to get R_{eq} , C_{eq} , or L_{eq} .

$$\tau = R_{eq}C_{eq} \quad (\text{for RC}) \quad \text{or} \quad \tau = \frac{L_{eq}}{R_{eq}} \quad (\text{for RL})$$

- 4. Complete Response:** The solution is:

$$f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-\frac{t}{\tau}} \quad \text{for } t \geq 0$$

Remarks

- $$f(t) = \underbrace{f(\infty)}_{\text{forced response}} + \underbrace{[f(0^+) - f(\infty)]}_{\text{natural response}} e^{-\frac{t}{\tau}}$$

- R_{eq} is the Thevenin resistance seen by the capacitor or inductor, after deactivating all independent sources.
- If a switch changed state (closes or opens) at $t = t_0$, then

$$v_C(t_0^+) = v_C(t_0^-)$$

"The voltage across a capacitor cannot change instantaneously."

$$i_L(t_0^+) = i_L(t_0^-)$$

"The current through an inductor cannot change instantaneously."

Note: Only state variables (v_C and i_L) are strictly continuous across switching boundaries. All other voltages and currents (v_R , v_L , i_C , i_R) can change instantaneously.

Source-Free RL and RC Networks

The solution of a homogeneous differential equation is a **transient response** of exponential form.

Source-Free RL Network

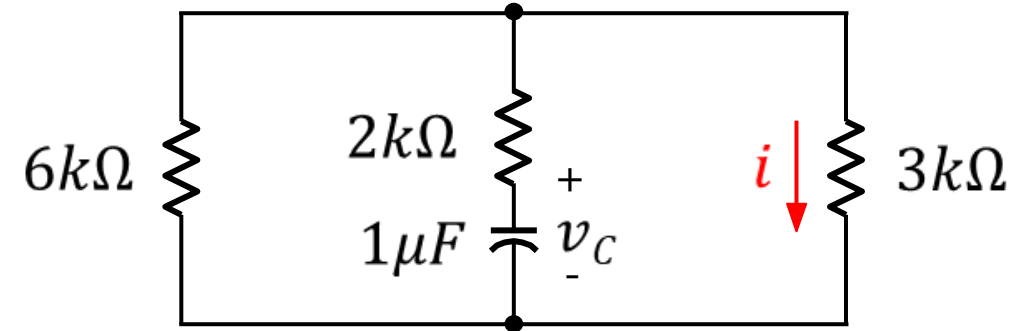
$$i_L(t) = K e^{-\frac{R}{L}t}$$

Source-Free RC Network

$$v_C(t) = K e^{-\frac{1}{RC}t}$$

Example 3

The circuit shown has several resistors but only one capacitor. Given $v_c(0^+) = 20$ V, find i for $t \geq 0$.



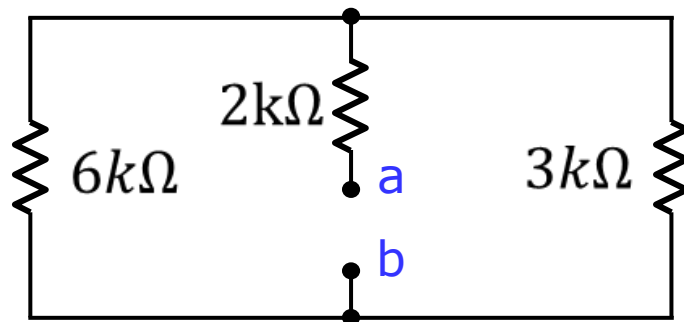
SOLUTION:

Initial Condition: It is given that at $t = 0^+$, $v_c(0^+) = 20$ V.

Final Condition: Since the circuit is source-free, the capacitor voltage will eventually become zero. Hence, $v_c(\infty) = 0$.

Example 3

Time Constant: Determine the equivalent resistance seen by the capacitor and use it to solve the time constant.



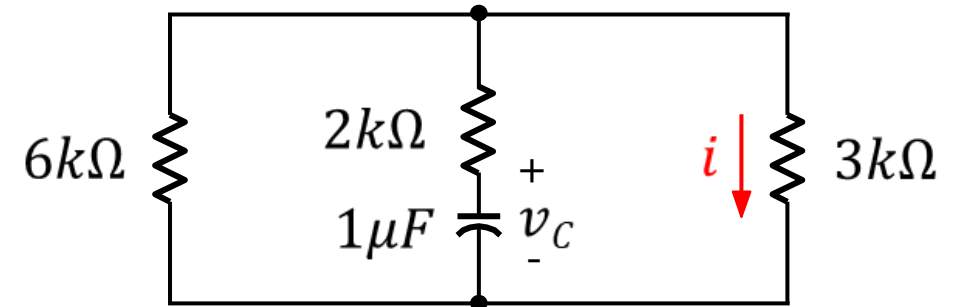
$$R_{ab} = 2k\Omega + \frac{6k\Omega(3k\Omega)}{6k\Omega + 3k\Omega} \\ = 4k\Omega$$

$$\tau = R_{ab}C = (4k\Omega)(1\mu F) = \frac{1}{250} \text{ s} \quad \text{or} \quad 4 \text{ ms}$$

Complete Response: Using the general expression $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-\frac{t}{\tau}}$, the capacitor voltage is

$$v_c(t) = 20e^{-250t} \text{ V} \quad t \geq 0$$

Example 3



Solve for $i(t)$.

- Using the equation $i_c(t) = C \frac{dv}{dt}$, the capacitor current is

$$i_c(t) = -5e^{-250t} \text{ mA} \quad t \geq 0$$

- Solve for $i(t)$ using current division.

$$i(t) = \frac{6 \text{ k}\Omega}{6 \text{ k}\Omega + 3 \text{ k}\Omega} (-i_c)$$

$$\boxed{i(t) = 3.33e^{-250t} \text{ mA} \quad t \geq 0}$$

RL and RC Networks with Constant Source

The solution of a non-homogeneous differential equation consists of two components: the **transient response** and the **steady-state response**.

RL Network with Constant Source

$$i_L(t) = A + Ke^{-\frac{R}{L}t}$$

↑ ↑
steady-state transient
response response

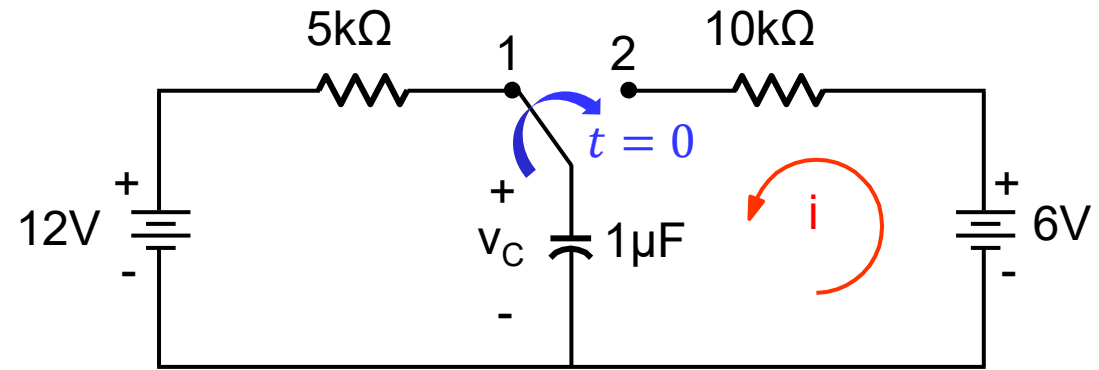
RC Network with Constant Source

$$v_C(t) = A + Ke^{-\frac{1}{RC}t}$$

↑ ↑
steady-state transient
response response

Example 4

The switch has been in position 1 for a long time. At $t = 0$, the switch is moved to position 2. Find the current $i(t)$ for $t \geq 0$.



SOLUTION:

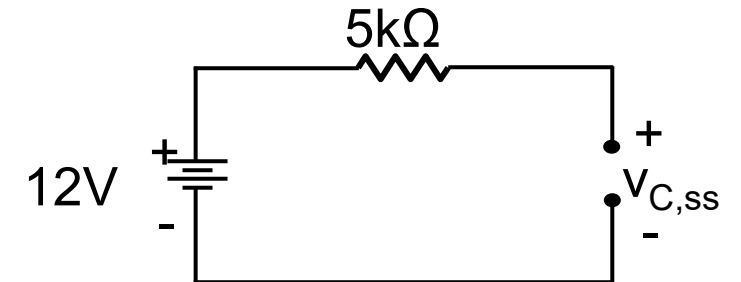
Initial Condition

Before switching, the circuit is in steady state, thus,

$$v_c(0^+) = v_c(0^-) = 12 \text{ V}$$

Use the $v_c(0^+)$ to get the initial current i .

$$i(0^+) = \frac{6 \text{ V} - 12 \text{ V}}{10 \text{ k}\Omega} = -0.6 \text{ mA}$$

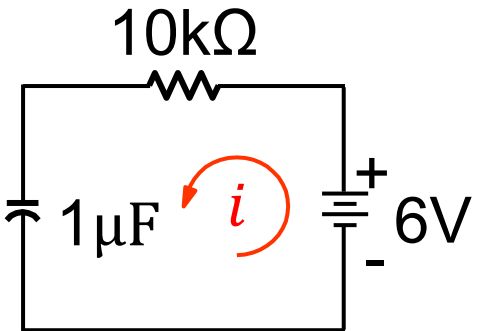


Equivalent circuit for $t < 0$.

Example 4

Final Conditions

The steady-state current is zero since the capacitor will be open-circuited when it reaches 6V. Thus, $i(\infty) = 0$.



Time Constant

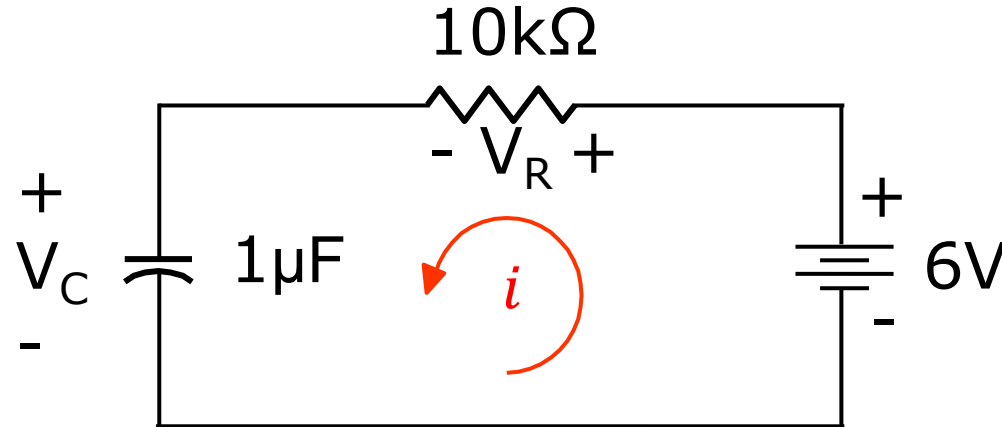
$$\tau = RC = 10\text{k}\Omega(1\mu\text{F}) = \frac{1}{100} \text{ s}$$

Finally, the expression describing i is described below:

$$i(t) = 0 + [-0.6 - 0]e^{-100t} \text{ mA}$$

$$i(t) = -0.6e^{-100t} \text{ mA} \quad t \geq 0$$

Comments



1. The actual current flows in the clockwise direction. The capacitor supplies the current. The 6-volt source is absorbing power.
2. The voltages across the resistor and capacitor can be found to be

$$v_R = Ri(t) = -6e^{-100t} \text{ V} \quad t \geq 0$$

$$v_C = 6 - v_R = 6 + 6e^{-100t} \text{ V} \quad t \geq 0$$

Comments

3. The energy stored in the capacitor decreases from 72 μJ to 18 μJ .

$$W_c(0^+) = \frac{1}{2} C v_c^2(0^+) = \frac{1}{2} (1\mu\text{F})(12\text{V})^2 = 72 \mu\text{J}$$

$$W_c(\infty) = \frac{1}{2} C v_c^2(\infty) = \frac{1}{2} (1\mu\text{F})(6\text{V})^2 = 18 \mu\text{J}$$

4. The resistor will dissipate a total energy of 18 μJ .

$$W_R = \int_0^\infty \frac{v_R^2}{R} dt = \int_0^\infty \frac{(-6e^{-100t})^2}{10\text{k}} dt = \int_0^\infty \frac{36}{10\text{k}} e^{-200t} dt = 18 \mu\text{J}$$

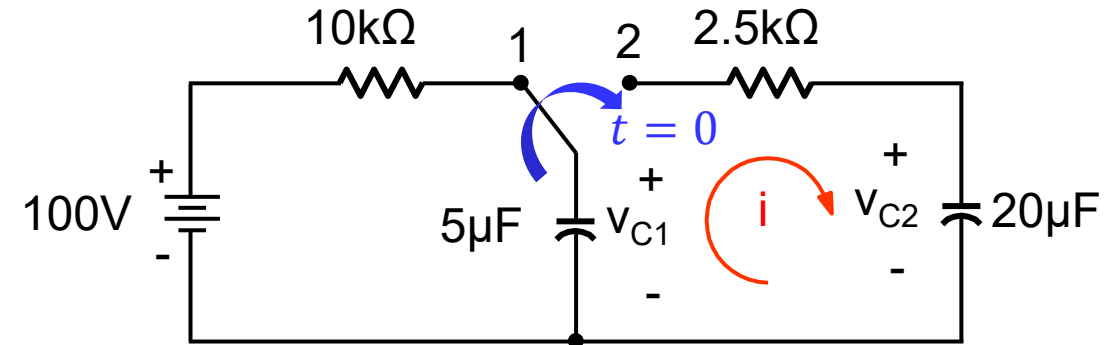
5. The 6V source will absorb a total energy of 36 μJ .

$$W_s = \int_0^\infty V_s i dt = \int_0^\infty 6 (-0.6e^{-100t} \text{ m}) dt = -36 \mu\text{J}$$

More Examples

Example 5

The network has reached steady-state condition with the switch in position 1. At $t = 0$, the switch is moved to position 2. Find i , v_{c1} and v_{c2} for $t \geq 0$. Assume that capacitor C_2 is initially uncharged.



Initial Conditions

The circuit is at steady-state prior to switching. Thus,

$$v_{c1}(0^+) = v_{c1}(0^-) = 100 \text{ V}$$

Also, since C_2 is initially uncharged,

$$v_{c2}(0^+) = v_{c2}(0^-) = 0$$

Then, the initial value of i is

$$i(0^+) = \frac{100\text{V} - 0\text{V}}{2.5\text{k}\Omega} = 40 \text{ mA}$$

Example 5

Final Conditions

By conservation of charge,

$$q_1 + q_2 = q_1' + q_2'$$

$$C_1 v_{c1}(0^+) + C_2 v_{c2}(0^+) = C_1 v_c(\infty) + C_2 v_c(\infty)$$

$$5\mu\text{F}(100\text{V}) + 20\mu\text{F}(0\text{V}) = 5\mu\text{F}[v_c(\infty)] + 20\mu\text{F}[v_c(\infty)]$$

$$\frac{500\mu\text{C}}{5\mu\text{F} + 20\mu\text{F}} = v_c(\infty)$$

$$v_c(\infty) = 20\text{V}$$

Note: The final voltages of C_1 and C_2 becomes equal and the current becomes $i(\infty) = 0$.

Example 5

Time Constant

$$C_{eq} = 4 \mu\text{F}$$
$$\tau = RC_{eq} = 2.5\text{k}\Omega(4\mu\text{F}) = \frac{1}{100} \text{ s or } 10\text{ms}$$

Finally, the expressions for i , v_{c1} , and v_{c2} are as follows

$$i(t) = 0 + [40 - 0]e^{-100t} \text{ mA}$$
$$v_{c1}(t) = 20 + [100 - 20]e^{-100t} \text{ V}$$
$$v_{c2}(t) = 20 + [0 - 20]e^{-100t} \text{ V}$$

$$\begin{aligned} i(t) &= 40e^{-100t} \text{ mA} & t \geq 0 \\ v_{c1}(t) &= 20 + 80e^{-100t} \text{ V} & t \geq 0 \\ v_{c2}(t) &= 20 - 20e^{-100t} \text{ V} & t \geq 0 \end{aligned}$$

Example 5

Alternative Solution

We can solve for $v_{c1}(t)$ and $v_{c2}(t)$ using the VI relationship in capacitors. This way, the conservation of charge is not used.

First, we can easily determine that $i(t) = 40e^{-100t}$ mA. For C_1 , we consider the current going in, which is $-i(t)$. Then,

$$\begin{aligned} v_{c1}(t) &= \frac{1}{C_1} \int_{-\infty}^t i(t) dt \\ &= v_{c1}(0^+) + \frac{1}{C_1} \int_0^t -i(t) dt \\ &= 100 + \frac{1}{5\mu\text{F}} \int_0^t (-40e^{-100t} \text{ mA}) dt \\ &= 100 + [80(e^{-100t} - 1)] \text{ V} \\ v_{c1}(t) &= \mathbf{20 + 80e^{-100t} \text{ V} \quad t \geq 0} \end{aligned}$$

Apply the same process to get $v_{c2}(t)$.

Comments

1. The current decays to zero but v_{c1} and v_{c2} do not decay to zero. At steady-state ($t = \infty$),

$$v_{c1,ss} = v_{c2,ss} = 20 \text{ V}$$

2. The initial energy stored in C_1 and C_2

$$W_{c1}(0^+) = \frac{1}{2} C_1 v_{c1}^2(0^+) = 25 \text{ mJ}$$

$$W_{c2}(0^+) = \frac{1}{2} C_2 v_{c2}^2(0^+) = 0$$

Comments

3. The final energy stored in C_1 and C_2

$$w_{c1}(\infty) = \frac{1}{2} C_1 v_{c1}^2(\infty) = 1 \text{ mJ}$$

$$w_{c2}(\infty) = \frac{1}{2} C_2 v_{c2}^2(\infty) = 4 \text{ mJ}$$

4. The total energy lost is 20 mJ.

5. The total energy dissipated by the resistor

$$W_R = \int_0^{\infty} i^2 R dt = \int_0^{\infty} 4 e^{-200t} dt = 20 \text{ mJ}$$

Note: At $t = 0^+$, $v_{c1} = 100$ volts and $v_{c2} = 0$. Capacitor C_1 supplies the current that charges capacitor C_2 . The current stops when $v_{c1} = v_{c2} = 20$ V.

Example 6

In the circuit, $v_{C1}(0^-) = 12 \text{ V}$ and $v_{C2}(0^-) = 0 \text{ V}$.

Find $i_R(t)$, $v_{C1}(t)$, and $v_{C2}(t)$.

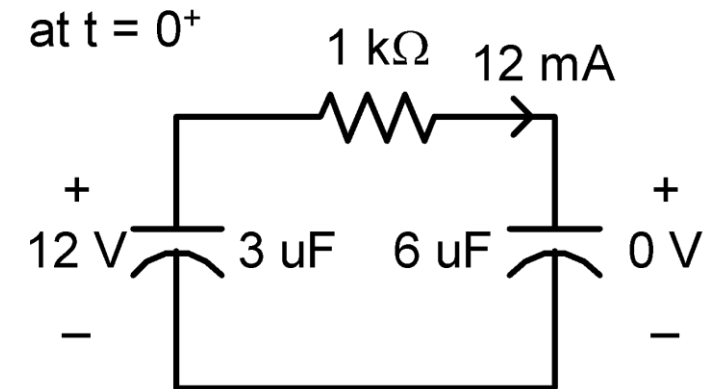
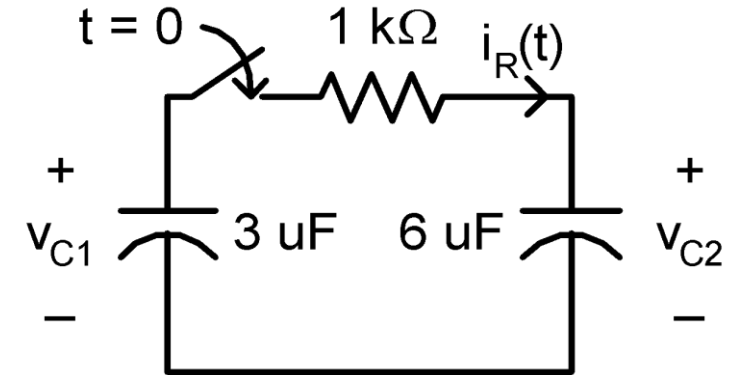
Initial Conditions

At $t = 0^+$:

$$v_{C1}(0^+) = v_{C1}(0^-) = 12 \text{ V}$$

$$v_{C2}(0^+) = v_{C2}(0^-) = 0 \text{ V}$$

$$i_R(0^+) = \frac{v_{C1}(0^+) - v_{C2}(0^+)}{1 \text{ k}\Omega} = \frac{12 \text{ V} - 0 \text{ V}}{1 \text{ k}\Omega} = 12 \text{ mA}$$



Example 6

Final Conditions

After a very long time, $i_R(\infty) = 0$. Hence,

$$v_{C1}(\infty) = v_{C2}(\infty)$$

$$\frac{q_1}{c_1} = \frac{q_2}{c_2} \rightarrow \frac{q_1}{3 \mu\text{F}} = \frac{q_2}{6 \mu\text{F}} \rightarrow 2q_1 = q_2$$

Initial charge stored = Final charge stored

$$12 \text{ V} (3 \mu\text{F}) = q_1 + q_2$$

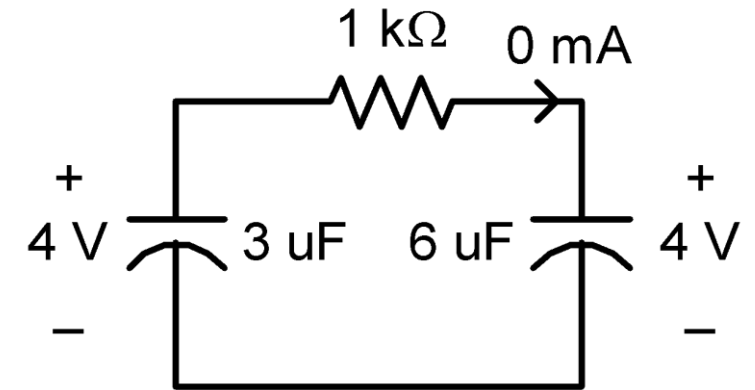
$$36 \mu\text{C} = q_1 + 2q_1$$

$$q_1 = 12 \mu\text{C}$$

$$q_2 = 24 \mu\text{C}$$

Therefore,

$$v_{C1}(\infty) = v_{C2}(\infty) = \frac{q_1}{c_1} = \frac{12 \mu\text{C}}{3 \mu\text{F}} = 4\text{V}$$



Example 6

Time Constant

$$R_{eq} = 1 \text{ k}\Omega$$

$$C_{eq} = \frac{3 \text{ }\mu\text{F} (6 \text{ }\mu\text{F})}{3 \text{ }\mu\text{F} + 6 \text{ }\mu\text{F}} = 2 \text{ }\mu\text{F}$$

$$\tau = R_{eq}C_{eq} = 1 \text{ k}\Omega(2 \text{ }\mu\text{F}) = \frac{1}{500} \text{ s} = 2 \text{ ms}$$

Finally, following the form $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-\frac{t}{\tau}}$

$$i_R(t) = 0 + [12 - 0]e^{-500t} \text{ mA}$$

$$i_R(t) = 12e^{-500t} \text{ mA} \quad t \geq 0$$

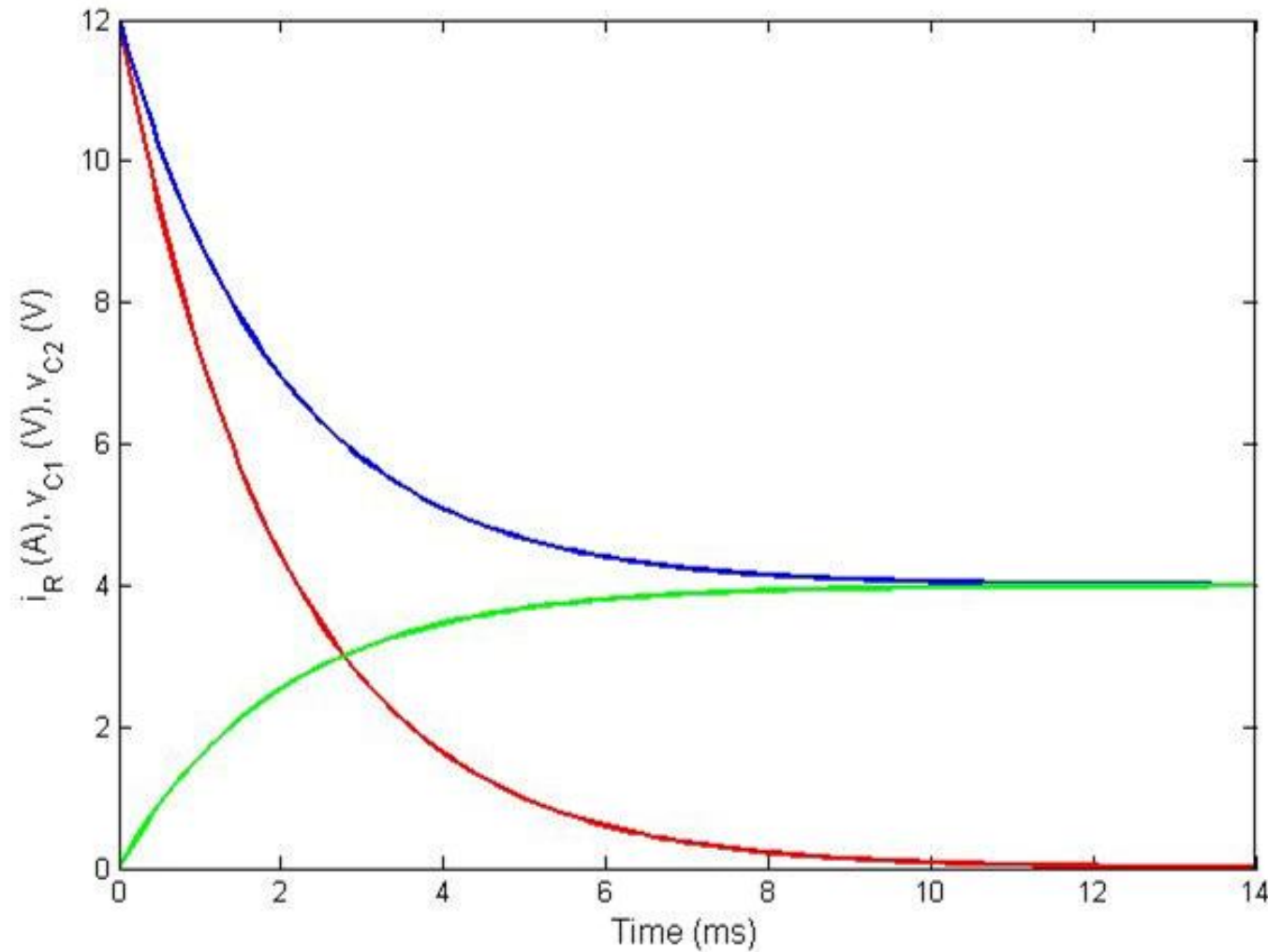
$$v_{c1}(t) = 4 + [12 - 4]e^{-500t} \text{ V}$$

$$v_{c1}(t) = 4 + 8e^{-500t} \text{ V} \quad t \geq 0$$

$$v_{c2}(t) = 4 + [0 - 4]e^{-500t} \text{ V}$$

$$v_{c2}(t) = 4 - 4e^{-500t} \text{ V} \quad t \geq 0$$

Example 6



$$i_R(t) = 12e^{-500t} \text{ mA} \quad t \geq 0$$

$$v_{C1}(t) = 4 + 8e^{-500t} \text{ V} \quad t \geq 0$$

$$v_{C2}(t) = 4 - 4e^{-500t} \text{ V} \quad t \geq 0$$

Example 7

Find the inductor current $i_L(t)$ and the inductor voltage $v_L(t)$.

Initial Conditions

$$i_L(0^+) = i_L(0^-) = 0$$

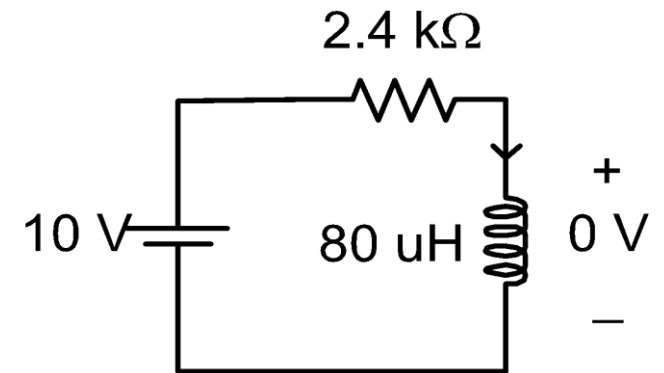
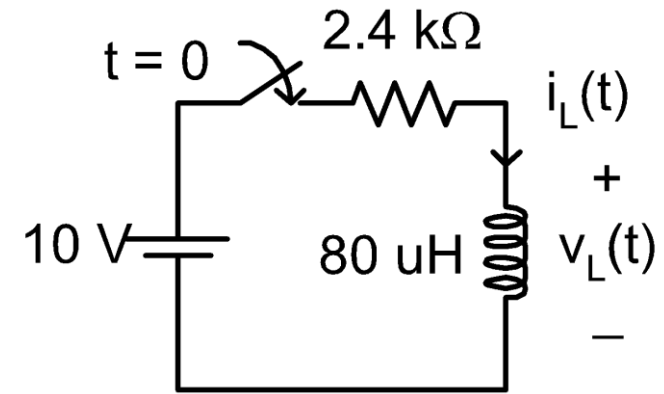
$$v_L(0^+) = 10 \text{ V}$$

Final Conditions

The inductor will behave like a short circuit, so

$$i_L(\infty) = \frac{10 \text{ V}}{2.4 \text{ k}\Omega} = 4.127 \text{ mA}$$

$$v_L(\infty) = 0 \text{ V}$$



Example 7

Time Constant

$$\tau = \frac{L_{eq}}{R_{eq}} = \frac{80 \mu\text{H}}{2.4 \text{ k}\Omega} = 33.33 \text{ ns}$$

Finally, following the form $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-\frac{t}{\tau}}$

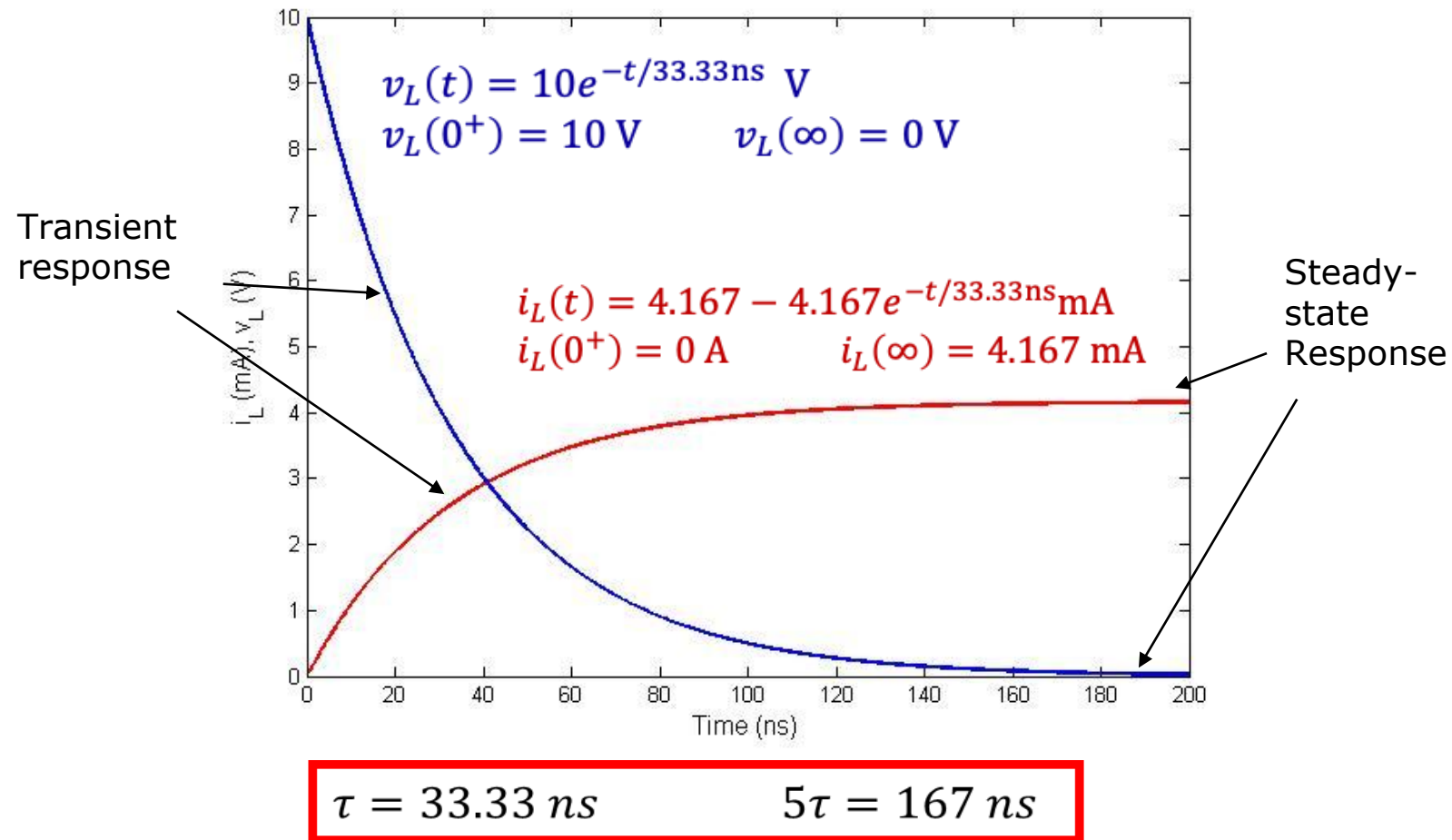
$$i_L(t) = 4.167 + [0 - 4.167]e^{-\frac{t}{33.33 \text{ ns}}} \text{ mA}$$

$$i_L(t) = 4.167 - 4.167e^{-\frac{t}{33.33 \text{ ns}}} \text{ mA} \quad t \geq 0$$

$$v_L(t) = 0 + [10 - 0]e^{-\frac{t}{33.33 \text{ ns}}} \text{ V}$$

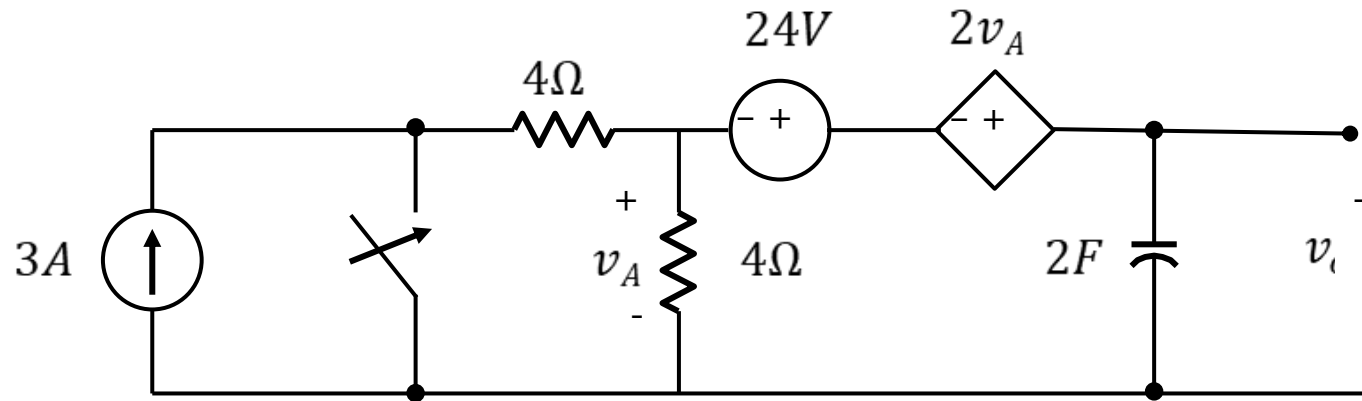
$$v_L(t) = 10e^{-\frac{t}{33.33 \text{ ns}}} \text{ V} \quad t \geq 0$$

Example 7



Example 8

If the switch in the network closes at $t = 0$, find $v_o(t)$ for $t > 0$.



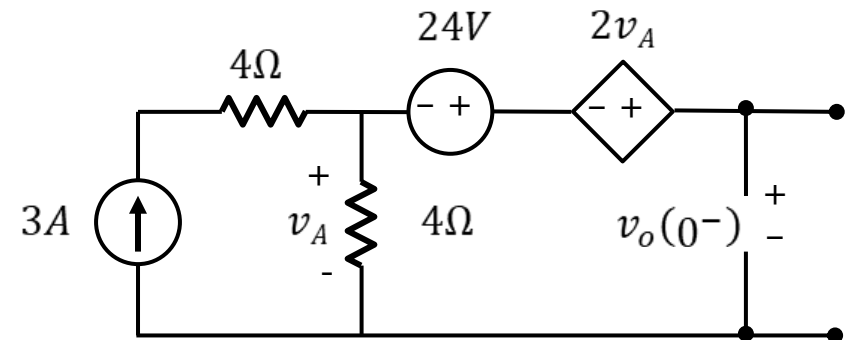
Initial Conditions

$$v_A(0^-) = 3A(4\Omega) = 12V$$

$$v_c(0^-) = 2v_A + 24 + v_A = 60V$$

Hence,

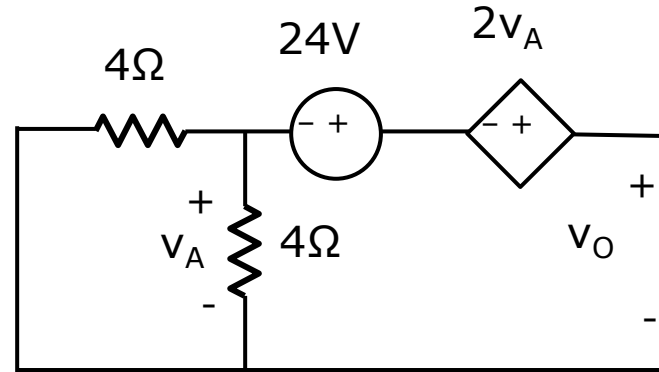
$$v_o(t) = v_c(0^+) = v_c(0^-) = 60V$$



Example 8

Final Conditions

$$v_A(\infty) = 0 \text{ V}$$
$$v_o(\infty) = 24 \text{ V}$$



Time Constant

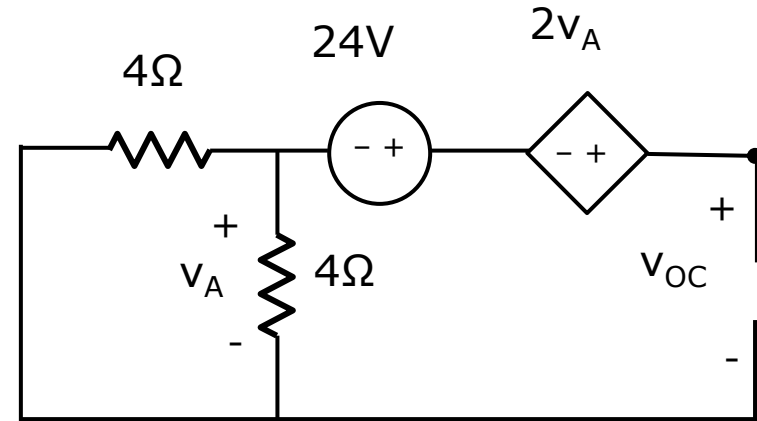
Since we have a dependent source, the equivalent resistance seen by the capacitor can be obtained by finding $R_{eq} = \frac{v_{oc}}{i_{sc}}$.

Example 8

Time Constant

Determine v_{OC} .

$$v_{OC} = 24 \text{ V}$$



Determine i_{SC} . By KVL,

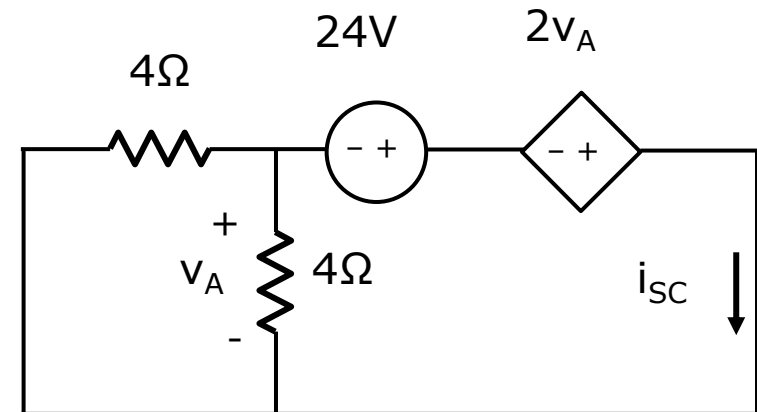
$$24 + 2v_A + v_A = 0$$

$$v_A = -8 \text{ V}$$

The two resistors are in parallel, thus

$$24 + 2v_A - 2i_{SC} = 0$$

$$i_{SC} = 4 \text{ A}$$



Example 8

Time Constant

The equivalent resistance is

$$R_{eq} = \frac{v_{oc}}{i_{sc}} = \frac{24\text{V}}{4\text{A}} = 6\Omega$$

Hence, the time constant is

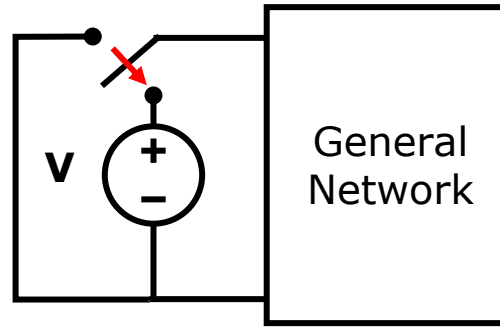
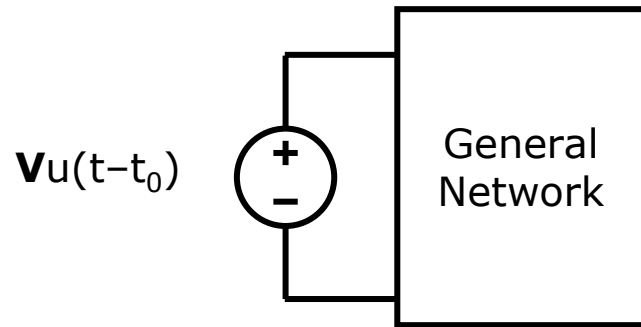
$$\tau = R_{eq}C = 6\Omega(2\text{F}) = 12 \text{ s}$$

Finally, following the form $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-\frac{t}{\tau}}$

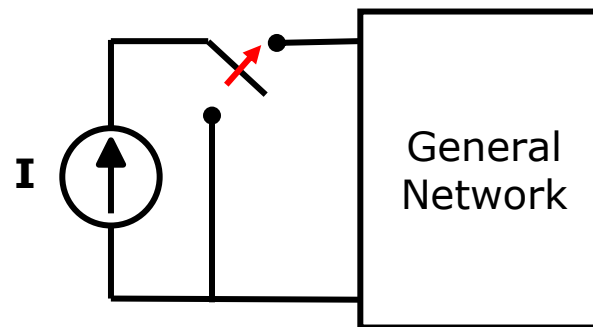
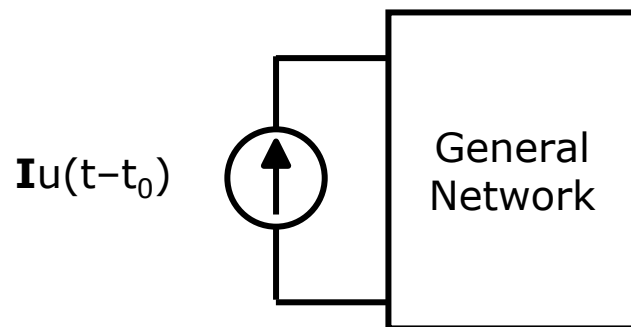
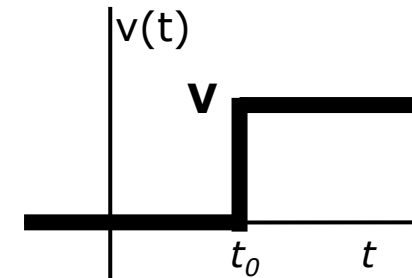
$$v_0(t) = 24 + [60 - 24]e^{-\frac{t}{12}} \text{ V}$$

$$v_0(t) = 24 + 36e^{-\frac{t}{12}} \text{ V} \quad t \geq 0$$

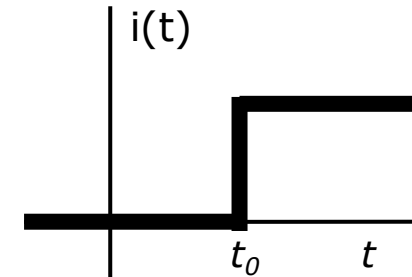
The Unit Step Function



Equivalent circuit

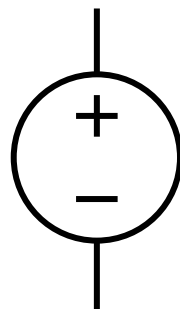


Equivalent circuit

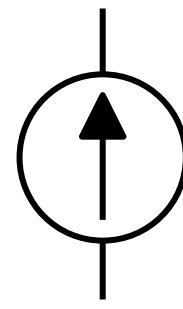


The Unit Step Function

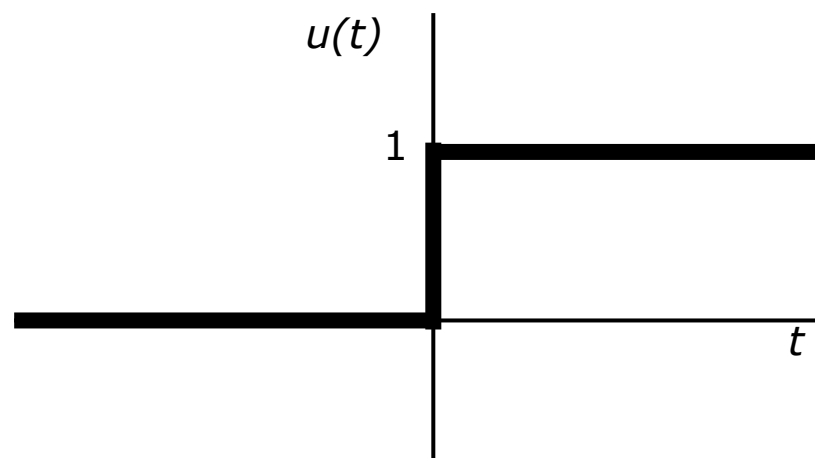
$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$



$u(t)$



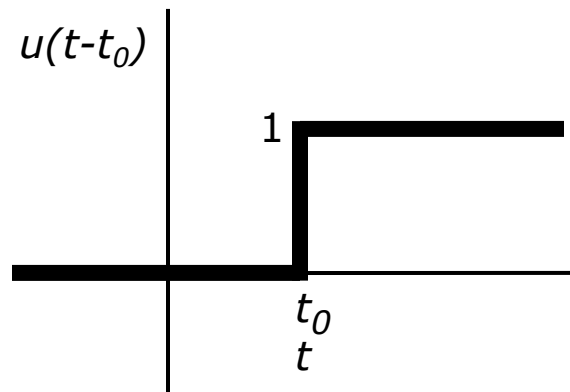
$u(t)$



The Unit Step Function

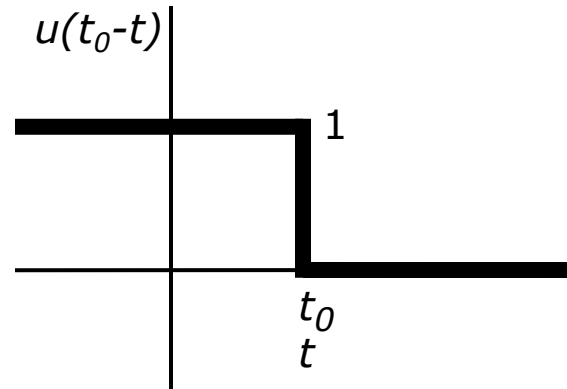
Translated Step Function

$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t \geq t_0 \end{cases}$$



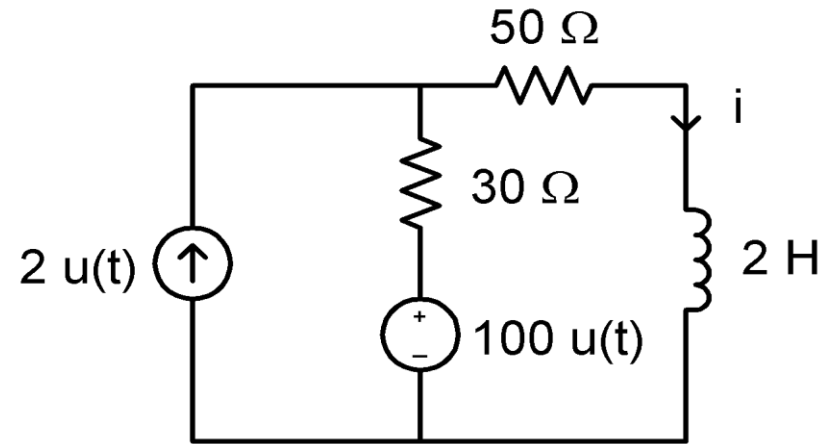
Step Function Inverted in Time

$$u(t_0 - t) = \begin{cases} 1, & t < t_0 \\ 0, & t \geq t_0 \end{cases}$$



Example 9

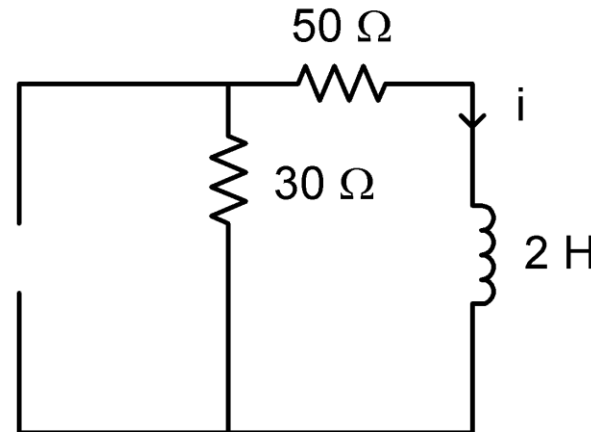
Find $i(t)$ for $t > 0$.



Initial Condition

When $t < 0$, the sources are off, thus,

$$i(0^+) = i(0^-) = 0 \text{ A}$$



Example 9

Final condition

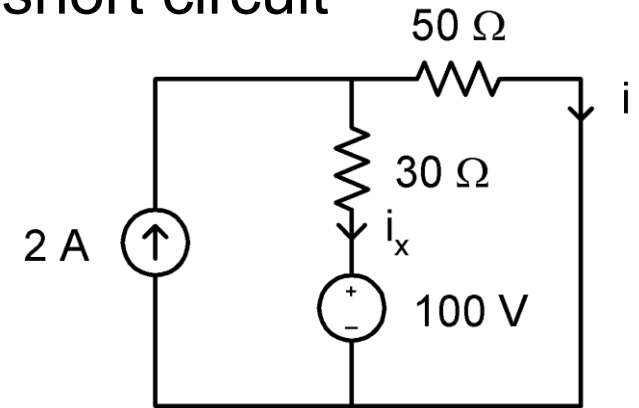
After a very long time, the inductor will behave like a short circuit

By KCL, $i + i_x = 2$.

By KVL, $-100 - 30i_x + 50i = 0$.

Solving the two equations give $i = 2$ A and $i_x = 0$.

Hence, $i(\infty) = 2$ A.

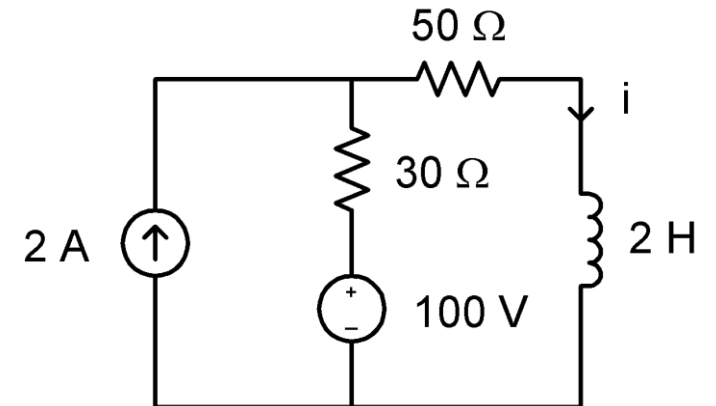


Time constant

$$L_{eq} = 2\text{H}$$

$$R_{eq} = 30\ \Omega + 50\ \Omega = 80\ \Omega$$

$$\tau = \frac{L_{eq}}{R_{eq}} = \frac{2\text{H}}{80\ \Omega} = \frac{1}{40}\text{ s or }25\text{ms}$$



Example 9

Finally, the expression for $i(t)$ is given by

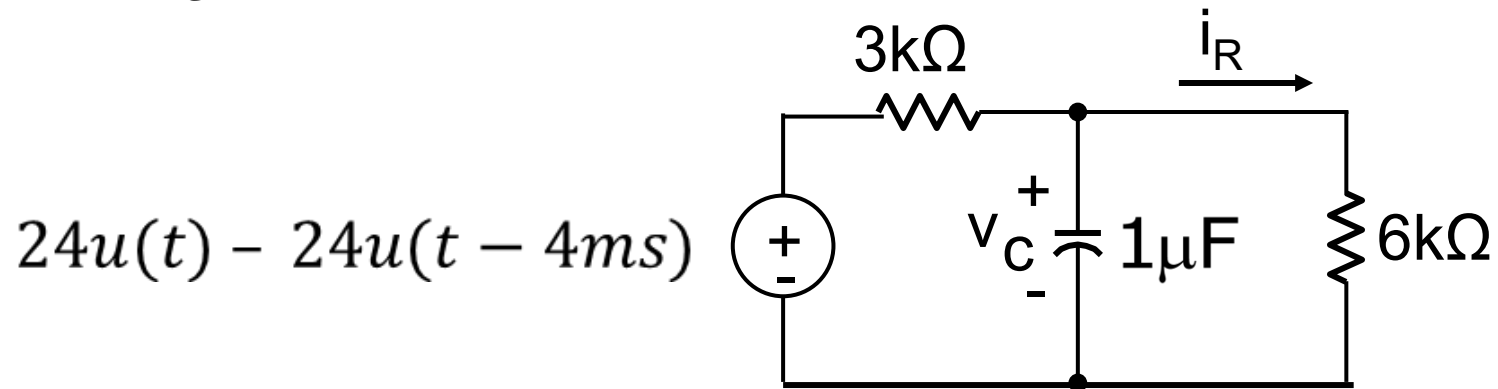
$$i(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-\frac{t}{\tau}}$$

$$i(t) = 2 + [0 - 2]e^{-40t} \text{ mA}$$

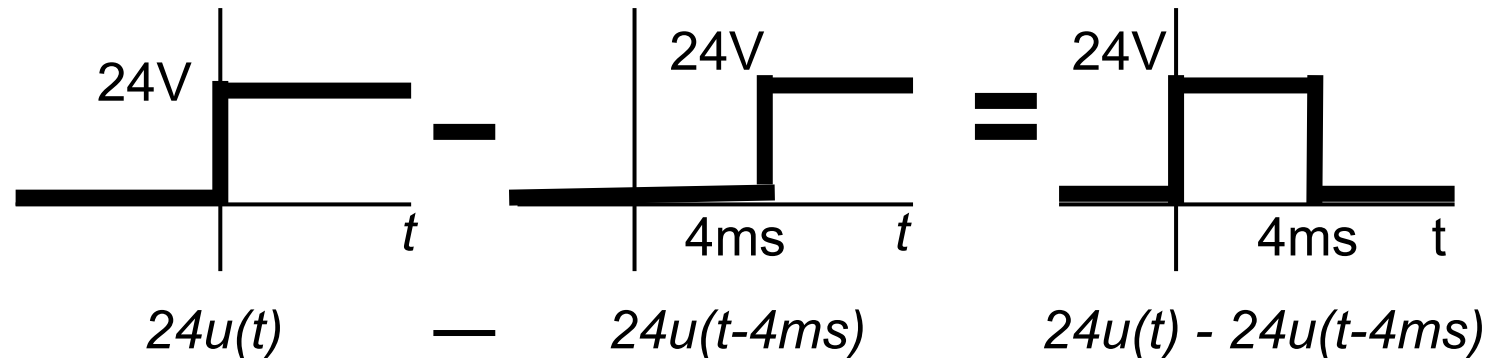
$$i(t) = 2 - 2e^{-40t} \text{ A} \quad t \geq 0$$

Example 10

The circuit shown is initially at steady-state condition. Formulate the expression for $v_C(t)$ and $i_R(t)$ for $t \geq 0$.



Evaluate the forcing function:



Example 10

We need to evaluate the circuit using two time intervals:

$0 < t < 4ms$, voltage source = 24V

$t > 4ms$, voltage source = 0

First time interval: $0 < t < 4ms$

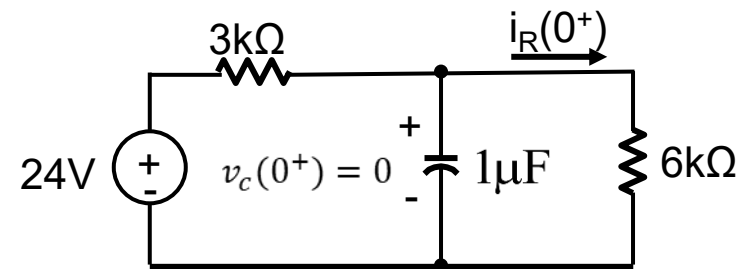
At $t < 0$, the circuit is in steady-state. The $3k\Omega$ and $6k\Omega$ resistors will dissipate whatever energy is initially stored in C, thus $v_C(0^-) = 0$.

Initial Conditions

$$v_C(0^+) = v_C(0^-) = 0$$

$$i_R(0^+) = 0$$

At $t = 0^+$

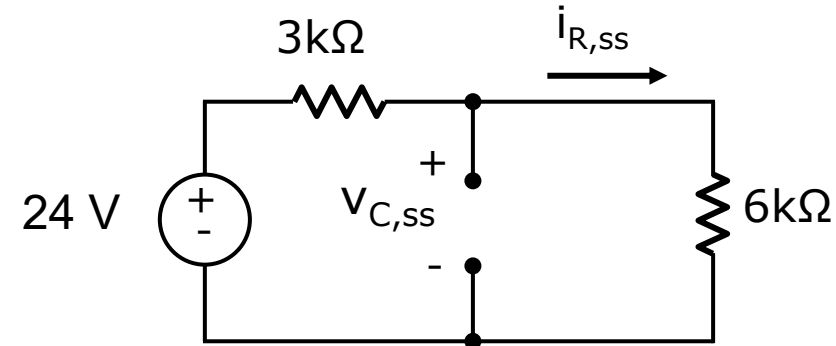


Example 10

Final Conditions

$$v_{C,ss} = (24V) \frac{6k\Omega}{3k\Omega + 6k\Omega} = 16V$$

$$i_{R,ss} = \frac{24V}{3k\Omega + 6k\Omega} = 2.67mA$$

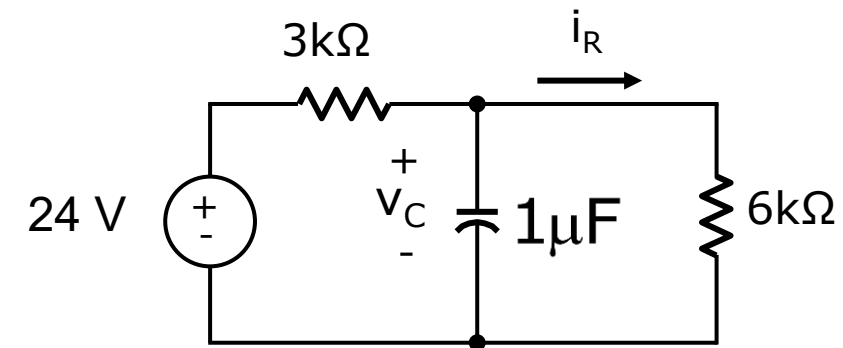


Time Constant

$$\tau = R_{eq}C$$

$$\tau = (2k\Omega)(1\mu F)$$

$$\tau = \frac{1}{500} \text{ s or } 2ms$$



Example 10

Complete Response $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-\frac{t}{\tau}}$

$$v_c(t) = 16 + [0 - 16]e^{-500t}$$

$$i_R(t) = 2.67 + [0 - 2.67]e^{-500t}$$

$$v_c(t) = 16 - 16e^{-500t} \text{ V} \quad 0 < t \leq 4\text{ms}$$

$$i_R(t) = 2.67 - 2.67e^{-500t} \text{ mA} \quad 0 < t \leq 4\text{ms}$$

Example 10

Second time interval: $t \geq 4ms$

Initial Conditions Determine the initial conditions from the expressions in the first time interval.

Right before switching,

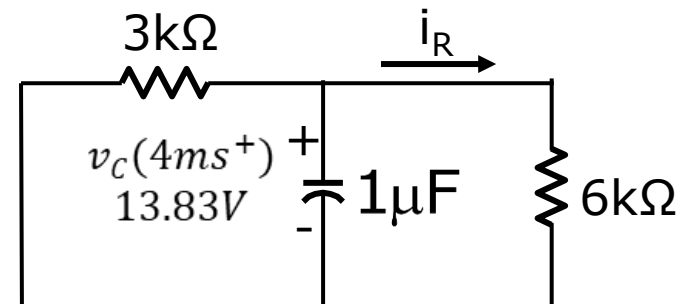
$$v_C(4ms^-) = 16 - 16e^{-500(4ms)} \text{ V} \approx 13.83 \text{ V}$$

$$i_R(4ms^-) = 2.67 - 2.67e^{-500(4ms)} \text{ mA} \approx 2.308 \text{ mA}$$

At $t = 4ms^+$

$$v_C(4ms^+) = v_C(4ms^-) = 13.83 \text{ V}$$

$$i_R(4ms^+) = i_R(4ms^{-1}) = 2.308 \text{ mA}$$



Example 10

Final Conditions This is a source-free network, so at steady-state $i'_{R,ss} = 0$ and $v'_{c,ss} = 0$.

Time Constant

$$R_{eq} = 3\text{k}\Omega \parallel 6\text{k}\Omega = 2\text{k}\Omega$$

$$\tau = R_{eq}C = 2\text{k}\Omega(1\mu\text{F}) = \frac{1}{500} \text{ s} = 2\text{ms}$$

Complete Response $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-\frac{t}{\tau}}$

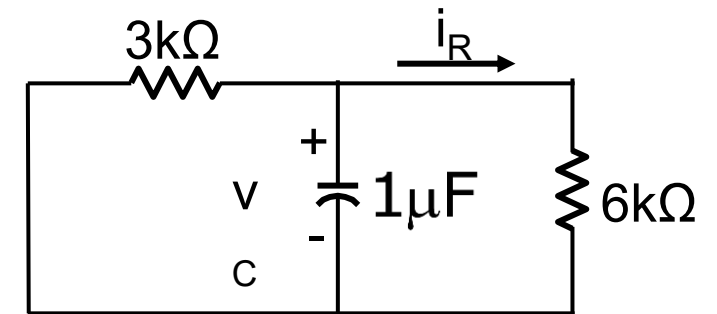
$$v_c(t) = 0 + [13.83 - 0]e^{-500(t-4\text{ms})}$$

$$i_R(t) = 0 + [2.308 - 0]e^{-500(t-4\text{ms})}$$

$$v_c(t) = 13.83e^{-500(t-4\text{ms})} \text{ V} \quad t > 4\text{ms}$$

$$i_R(t) = 2.308e^{-500(t-4\text{ms})} \text{ mA} \quad t > 4\text{ms}$$

Equivalent circuit for $t \geq 4 \text{ ms}$.



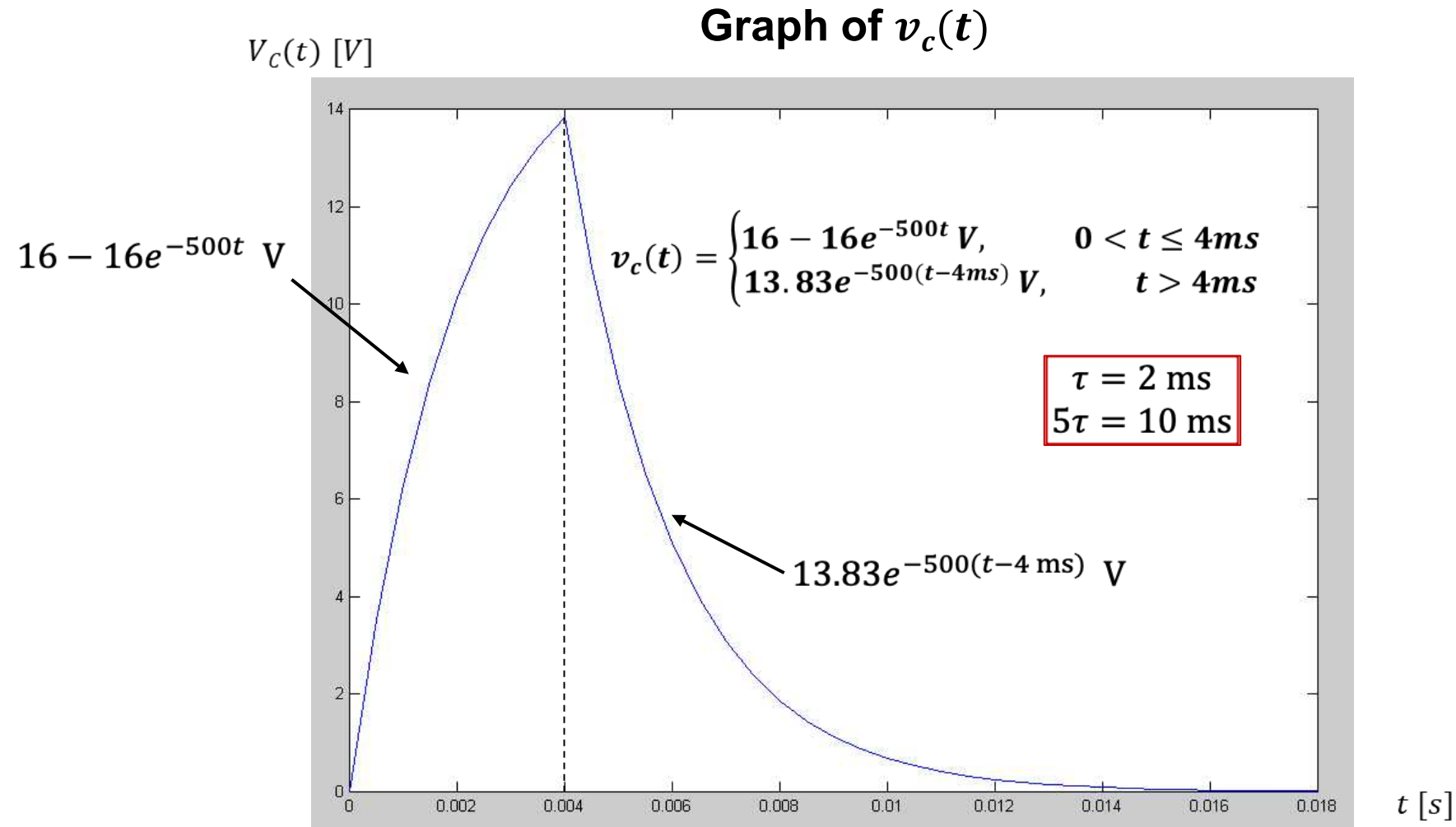
Example 10

Finally, combining the complete response of the first and second time intervals, we have

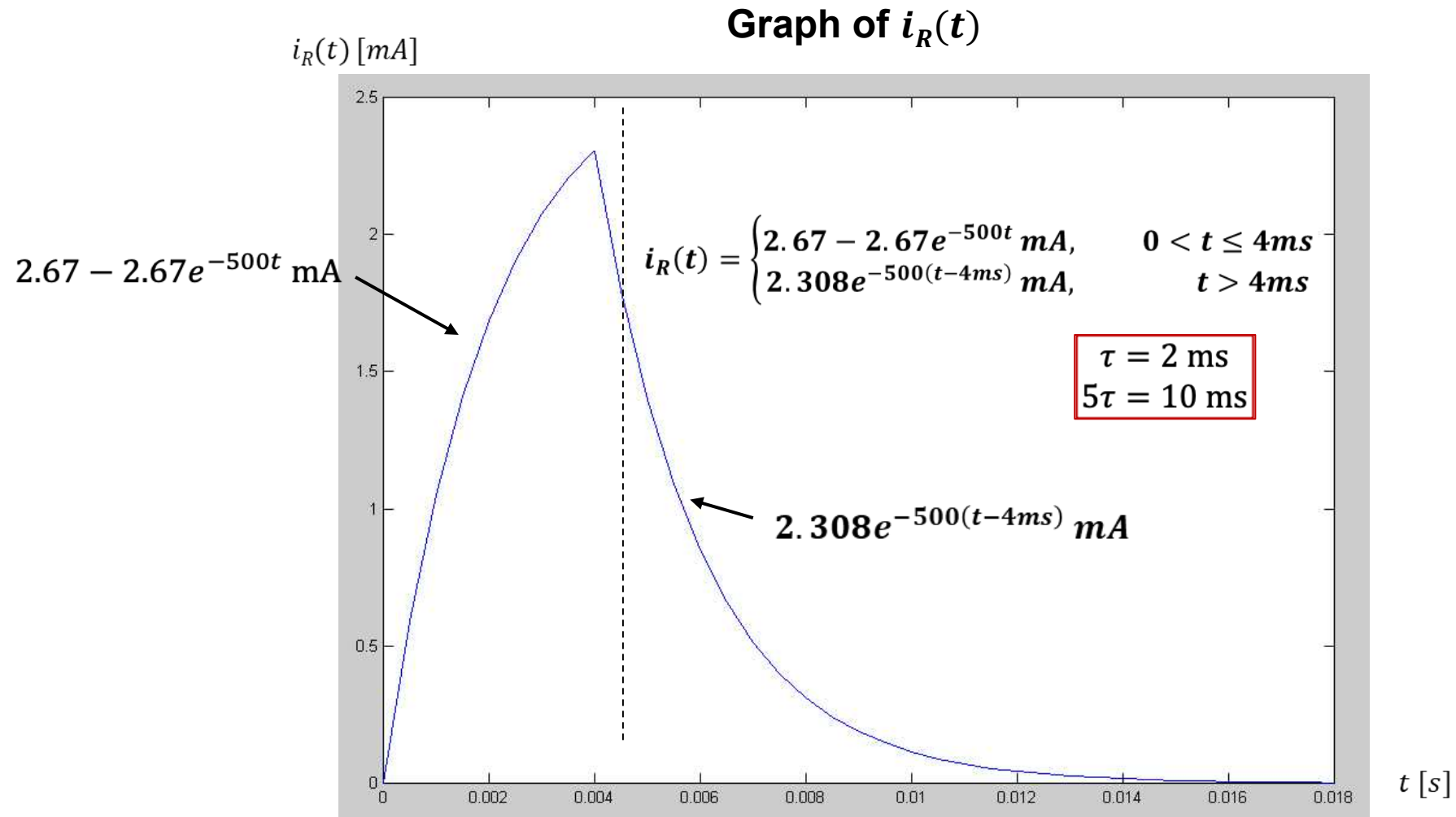
$$v_c(t) = \begin{cases} 16 - 16e^{-500t} \text{ V}, & 0 < t \leq 4\text{ms} \\ 13.83e^{-500(t-4\text{ms})} \text{ V}, & t > 4\text{ms} \end{cases}$$

$$i_R(t) = \begin{cases} 2.67 - 2.67e^{-500t} \text{ mA}, & 0 < t \leq 4\text{ms} \\ 2.308e^{-500(t-4\text{ms})} \text{ mA}, & t > 4\text{ms} \end{cases}$$

Example 10



Example 10



Summary

- For any first-order RL or RC circuit , the complete response $f(t)$ for $t \geq t_0$ is given by the general step formula:

$$f(t) = f(\infty) + [f(t_0^+) - f(\infty)]e^{-\frac{t-t_0}{\tau}}$$

- A unit step function $u(t)$ represents abrupt switching at $t = 0$:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

- Circuits excited by pulses are analyzed in consecutive time intervals ($0 < t < T$ and $t \geq T$) , where the final state of the first interval establishes the initial condition for the second interval.